An Intelligent Essay-Based Learning Support System with Difficulty Prediction and Feedback-Driven Improvement

Project ID: 25-26J-089

Project Proposal Report

Aaraby T – IT22598196

B.Sc. (Hons) Degree in Information Technology Specializing in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research proposes an intelligent, AI-driven, adaptive essay-based learning support system for students in the Database Management Systems (DMS) module at the Sri Lanka Institute of Information Technology (SLIIT). The system enhances assessment preparation and academic performance by leveraging Natural Language Processing (NLP), transformer-based models (BERT), reinforcement learning, AutoML, and Bloom's Taxonomy. It creates a dynamic, self-paced learning environment tailored to the DMS curriculum with core functionalities: predicting essay question difficulty, categorizing past exam questions, and evaluating student essays. Using past exam papers and lecture materials, the system generates a multi-level difficulty index (Easy, Medium, Hard) based on content alignment, linguistic complexity, and cognitive demands, enabling progressive practice. A repository of categorized past paper questions allows students to select questions matching their skill level through an intuitive interface with progress tracking. Automated essay evaluation employs semantic similarity measures and Bloom's Taxonomy criteria to assess responses for accuracy, coherence, and relevance, providing a satisfaction score and personalized feedback with targeted revision suggestions (e.g., query optimization, database normalization). Reinforcement learning and AutoML optimize feedback, adapting to individual performance trends to address learning gaps and recommend tailored practice. Designed for scalability, usability, and data privacy, the system's modular architecture supports future enhancements like real-time progress tracking or integration with learning management systems, with potential adaptation to other disciplines by retraining models. This AI-powered tool enhances student outcomes and assessment efficiency in technical education.

Keywords: Artificial Intelligence, Essay-Based Learning, Database Management Systems, Natural Language Processing, Transformer Models, Reinforcement Learning, AutoML, Bloom's Taxonomy, Adaptive Learning, Automated Assessment, Personalized Feedback

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1.0 INTRODUCTION

Assessment is one of the most influential drivers of student learning in higher education, yet many universities struggle to maintain consistent alignment between curriculum delivery, lecture content, and examination practices. This misalignment often results in students devoting significant time to topics that appear only rarely in exams, while overlooking those that are frequently tested and more critical for success. As a consequence, students' preparation becomes inefficient and learning outcomes are not fully achieved. Traditional approaches, such as manually analyzing past exam papers and lecture materials, are often subjective, inconsistent, and time-consuming, making them inadequate for addressing this issue on a larger scale.

The emergence of Artificial Intelligence (AI), Natural Language Processing (NLP), and educational analytics presents a timely opportunity to transform this process. Unlike conventional learning management systems, which largely focus on content delivery and administrative functions, AI-powered approaches can uncover structural and topical trends across years of academic assessments, identify gaps between taught material and examined content, and provide actionable insights for both learners and educators. These capabilities not only optimize revision strategies for students but also guide institutions in improving curriculum design and instructional effectiveness.

Our research targets the development of an intelligent, AI-powered academic support framework that systematically analyses historical exam papers, lecture content, and continuous assessments to deliver data-driven insights and personalized learning outputs. Through advanced NLP techniques, the system identifies high-frequency topics, predicts question difficulty, and classifies content according to syllabus requirements, while also highlighting redundancies and underrepresented areas. Based on these analyses, the framework generates practical learning aids, including model exam papers that reflect historical patterns, concise and curriculum-aligned short notes, adaptive essay evaluation with feedback, interactive flashcards, voice notes, and personalized time-aware study schedules.

By integrating these elements into a unified platform, the research provides a holistic solution that enhances student self-learning, improves exam preparation efficiency, and strengthens curriculum alignment with assessment practices. Ultimately, the project seeks to bridge the gap between academic delivery and evaluation through a data-driven, scalable, and personalized educational support system, thereby contributing to improved student achievement and institutional teaching quality.

1.1 Background & Literature Survey

The integration of Artificial Intelligence (AI) in educational assessment has significantly transformed higher education, particularly in technical disciplines like the Database Management Systems (DMS) module at the Sri Lanka Institute of Information Technology (SLIIT). In this context, essay-based assessments evaluate students' mastery of complex concepts, critical reasoning, and applied understanding. Traditional manual evaluation methods are often time-consuming, subjective, and inconsistent. However, recent advancements in AI, particularly Natural Language Processing (NLP) and transformer-based models like BERT and GPT, reinforcement learning, and AutoML, now enable automated evaluation, personalized feedback, and adaptive learning support.

Studies by Pariyanto and Tungka (2025) showed that tools like ChatGPT can provide formative feedback to enhance narrative essay writing for EFL students (1). Chen et al. (2024) proposed a multi-strategy EFL writing system using deep learning to deliver targeted feedback and improve outcome(2). These studies highlight the potential of AI in general writing contexts, but do not address the specific challenges in technical modules like DMS. Automated essay scoring has also gained attention, with Xu, Mahmud, and Hoo (2024) conducting a systematic review demonstrating the effectiveness of such systems in educational settings, while noting limitations in handling nuanced, domain-specific content such as DMS (3). Foundational work by Pérez Marín (2004) explored statistical and shallow NLP techniques for short essays, laying the groundwork for modern AI approache(4).

Additionally, research on cognitive assessment and difficulty prediction, such as the intelligent parametric assessment model by Javed, Aslam, and Khurshid (2025), emphasizes integrating difficulty prediction and feedback in e-learning, though it is not specifically tailored to essay-based assessments (5). Many systems still rely on post-hoc performance data rather than proactively predicting difficulty based on content. Incorporating Bloom's Taxonomy into AI-powered assessment allows for structured evaluation of cognitive complexity. Furthermore, adaptive systems can adjust to individual learners' needs, as demonstrated by Chen et al. (2024) (2). However, few systems proactively predict difficulty using lecture content or integrate reinforcement learning for dynamic adaptation, leaving gaps in the development of student-focused, domain-specific systems for DMS.

1.2 Research Gap

The integration of Artificial Intelligence (AI) in educational systems has made progress; however, substantial gaps remain when evaluated against the 12 identified features. As shown in the comparative analysis table, content-based approaches, difficulty prediction, and lecture–answer correlation analysis are absent across most studies, with only limited progress in [2] and [5], and none providing a complete solution. Multi-level difficulty indexing is inconsistently addressed, appearing only in [5] as a basic implementation. Similarly, practice question categorization is entirely unaddressed in all reviewed works.

While automated essay evaluation is explored in [1], [2], [3], and [4], these contributions lack integration with broader adaptive mechanisms. Semantic similarity analysis is present in [2], [3], [4], and [5], yet remains partial and fragmented in its application. Bloom's Taxonomy integration is almost entirely absent, with only [5] offering a basic use case. Personalized feedback is addressed in [1], [2], and [5], but structured topic revision recommendations are missing from the majority of studies, appearing only in [1] and [2]. The adaptive learning loop is implemented only at a basic level in [2], leaving significant room for improvement.

The proposed system addresses these gaps, integrating content-based difficulty prediction, lecture–answer correlation, and a robust multi-level difficulty index. It introduces practice question categorization, advanced essay evaluation with reinforcement learning and AutoML, and Bloom's Taxonomy integration, providing personalized feedback, targeted revision recommendations, and a dynamic adaptive learning loop for SLIIT's DMS students.

Feature	Research 01 [1]	Research 02 [2]	Research 03 [3]	Research 04 [4]	Research 05 [5]	Proposed System
Content-Based	✗	✓	✗	✗	✗	✓
Difficulty Prediction	✗	✗	✗	✗	✓	✓
Lecture-Answer Correlation Analysis	✗	✗	✗	✗	✗	✓
Multi-Level Difficulty Index	✗	✗	✗	✗	✓(Basic)	✓
Practice Question Categorization	✗	✗	✗	✗	✗	✓
Automated Essay Evaluation	✓	✓	✓	✓	✗	✓
Semantic Similarity Analysis	✗	✓	✓	✓	✓	✓
Bloom's Taxonomy Integration	✗	✗	✗	✗	✓	✓
Personalized Feedback	✓	✓	✗	✗	✓	✓
Topic Revision Recommendations	✓	✓	✗	✗	✗	✓
Adaptive Learning Loop	✗	✓(Basic)	✗	✗	✗	✓

Figure 1- Competitive Analysis

1.3 Research Problem

The development of an intelligent learning support system for the Database Management Systems (DMS) module at the Sri Lanka Institute of Information Technology (SLIIT) is driven by a complex array of challenges that significantly affect students.Students encounter substantial difficulties, including uncertainty about the complexity of essay questions, which undermines their ability to devise effective preparation strategies and build confidence in tackling assessments. This issue is compounded by the absence of structured, categorized practice frameworks, which prevents them from progressing systematically in developing critical skills such as database design or query optimization. Furthermore, the feedback they receive is often generic, lacking detailed and specific guidance for revision, leading to repeated errors and persistent gaps in understanding. A notable disconnect also exists, as students struggle to bridge the gap between the unstructured nature of lecture materials and the specific requirements of exam questions, leaving them ill-prepared for the demands of the curriculum.

This research addresses: How can an intelligent system predict essay question difficulty using past exam papers and lecture materials, provide automated, accurate evaluation, and deliver dynamic, personalized feedback using NLP, reinforcement learning, and AutoML to foster an adaptive learning environment for DMS students? The system aims to empower students with clear preparation pathways and tailored learning support.

2.0 OBJECTIVES

2.1 Main Objectives

The primary objective is to develop an intelligent, adaptive essay-based learning support system for SLIIT's DMS students using AI, NLP, reinforcement learning, and AutoML. The system will predict essay question difficulty using past exam papers and lecture materials, enabling structured preparation aligned with skill levels. It will evaluate responses using NLP for accuracy, coherence, and cognitive depth, delivering personalized feedback with revision suggestions. Reinforcement learning and AutoML will optimize feedback and adapt recommendations based on performance trends, fostering an adaptive learning environment that improves assessment performance.

2.2 Specific Objectives

1. Data Collection & Preprocessing:

- Collect and preprocess DMS past papers and lecture content to create a comprehensive and structured dataset for subsequent analysis and system development.

2. Essay Difficulty Prediction:

- Develop models to predict question difficulty using linguistic and cognitive features, enabling a nuanced understanding of content complexity and supporting tailored preparation strategies.

3. Automated Essay Scoring:

- Implement NLP-based scoring with semantic and rubric-based criteria to evaluate student responses accurately and consistently.

4. Learning Analytics:

- Provide performance analytics and personalized recommendations to support individual student progress and address specific learning needs.

5. System Integration:

- Develop user-friendly interfaces with progress tracking, difficulty filters, and question categorization, validated for accessibility.

3.0 METHODOLOGY

3.1 System description

The system integrates supervised machine learning, NLP, reinforcement learning, and AutoML tailored for SLIIT's DMS students. It uses transformer-based models (e.g., BERT) for difficulty prediction and essay evaluation, reinforcement learning to adapt feedback dynamically, and AutoML to optimize feedback accuracy. NLP enables semantic analysis and feedback generation, creating a robust, student-focused, adaptive learning framework.

3.2 Key Components

1. Data Collection & Preprocessing:

- **Sources:** Gather comprehensive sets of SLIIT DMS past exam papers, lecture content, and syllabus (PDF/DOCX), with academic replication approval.
- **Tools:** Text extraction, normalization, and error correction applied to construct a structured dataset.
- **Process:** Clean raw text, remove special characters, tokenize, and standardize terminology to ensure consistency.
- **Output:**High-quality dataset ready for analysis and model training.

2. Difficulty Prediction Module:

- **Method:** Fine-tuned transformer-based models (BERT) for predicting essay question difficulty.
- **Features:** Considers linguistic complexity, cognitive demand based on Bloom's Taxonomy, and alignment with lecture
- **Advantage:** Addresses limitations of generic difficulty prediction by correlating directly with questions.
- **Output:** Predicted difficulty levels with Bloom's Taxonomy.

3. Automated Evaluation Module:

- **Method:** Advanced NLP techniques (semantic similarity, Bloom's criteria-based scoring).
- **Evaluation:** Assess essays on content accuracy, coherence, and cognitive depth.
- **Validation:** Automated scoring compared against expert evaluation for consistency.
- **Output:** Reliable essay scores aligned with DMS learning outcomes.

4. Feedback Generation Module:

- **Method:** NLP, reinforcement learning, and AutoML for dynamic feedback generation.
- **Features:** Personalized revision suggestions, topic-specific recommendations, and adaptive learning pathways.
- **Goal:** Dynamic feedback loop adapting to performance trends for tailored learning.
- **Output:** Actionable feedback for students.

5. User Interface & Integration:

- **Framework:** Develop an interactive and responsive interface (e.g., Streamlit).
- **Features:** Progress tracking, difficulty filters, and integrated access to all modules.
- **Validation:** Iterative testing with SLIIT students for usability and adaptation.
- **Output:** Student-centered interface supporting seamless learning experience.

3.3 Tools & Technology

- Programming: Python (primary), JavaScript (front end).
- NLP & ML: PyTorch, TensorFlow, spaCy, scikit-learn, HuggingFace Transformers
- Reinforcement Learning & AutoML: Agent reinforcement trainer(GRPO)
- OCR: Tesseract OCR
- Databases: SQLite, Elasticsearch,MongoDB
- Web Front End: React.js; Tailwind CSS, Streamlit.
- Design/Diagrams: Figma; draw.io.
- IDE: VS Code, Google Colab, Jupyter Notebook, Kaggle Platform
- VCS: GitHub
- Collaboration: Microsoft Teams, Slack.
- Testing: pytest (Python), Postman (APIs), Jest (JS).
- Deployment: Docker, Kubernetes, Jenkins (CI/CD)
- Project Management: Jira

System Work Flow Diagram

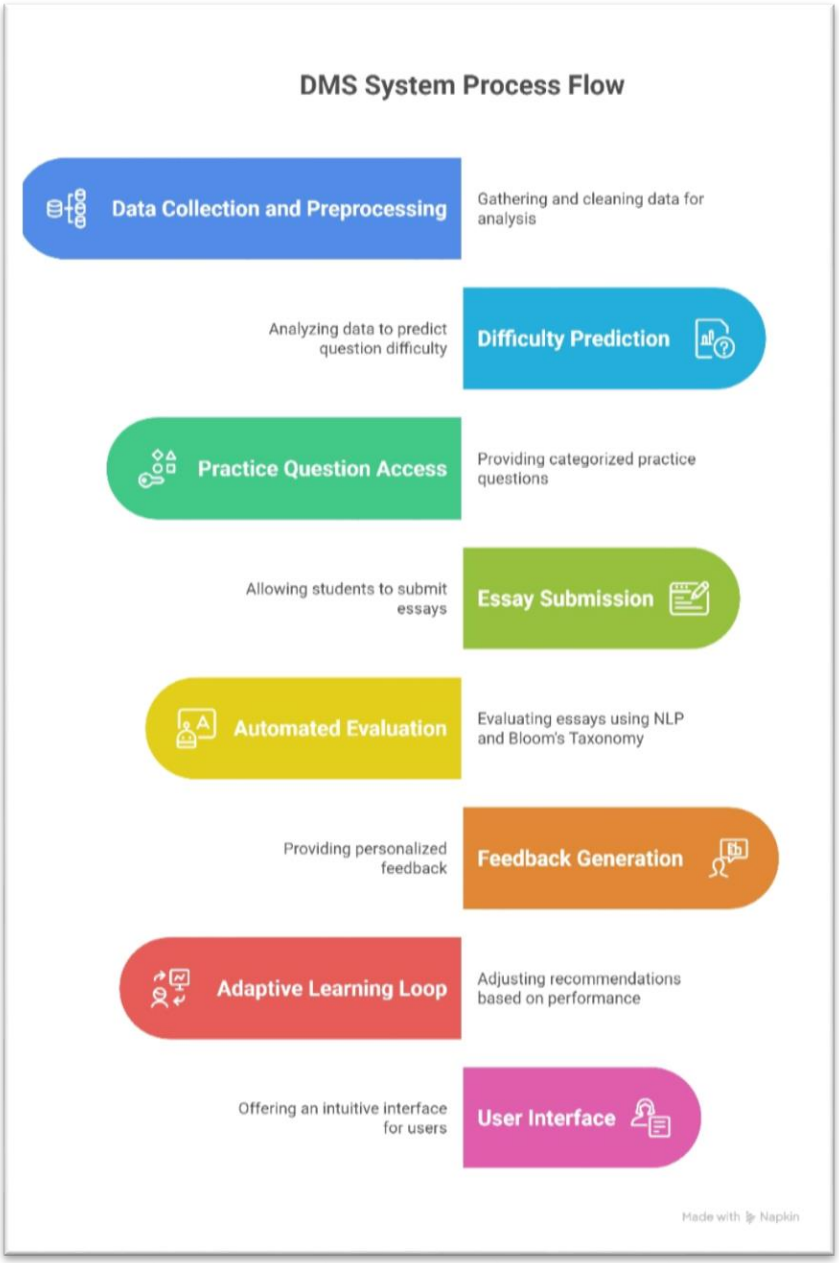

Figure 2: System Diagram

Overall system diagram

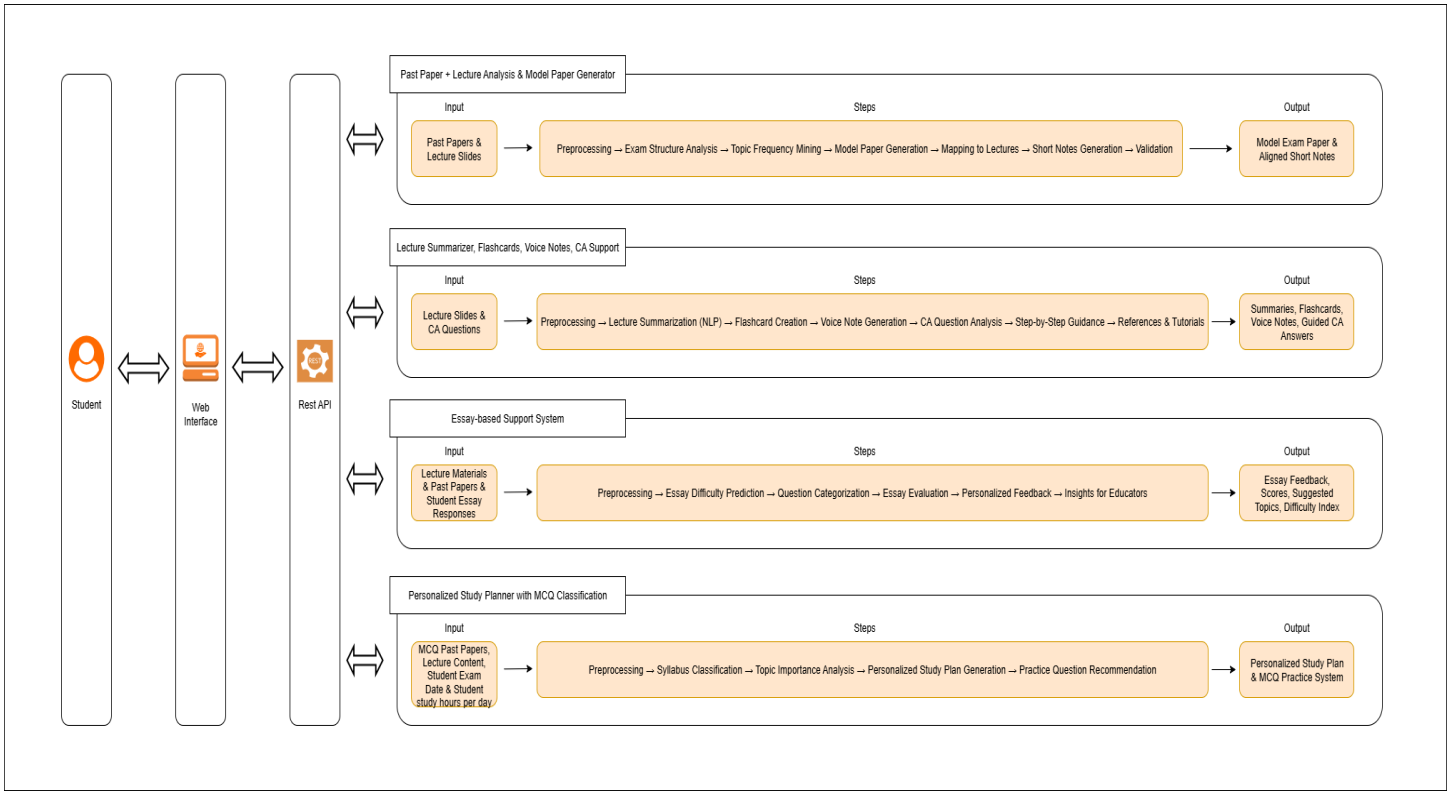

Figure 3: Overall System Diagram

Component Flow Diagram

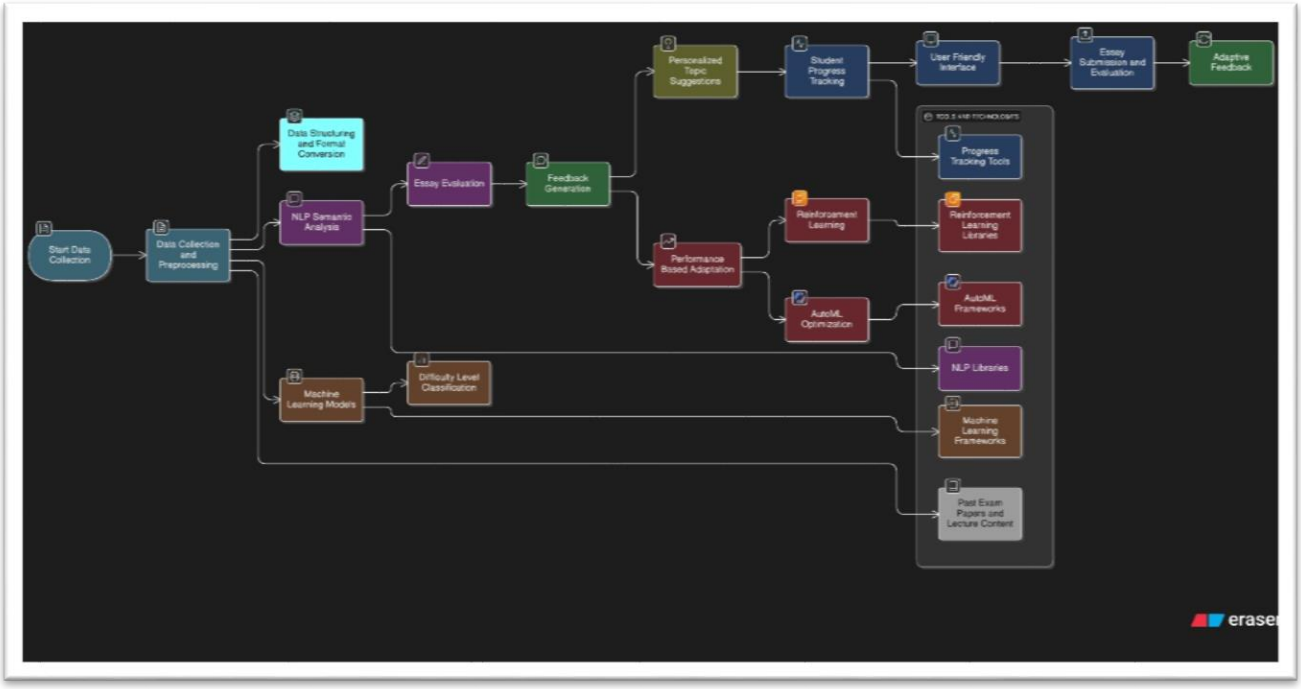

Figure 4: Component Diagram

3.4 PROJECT REQUIREMENTS

3.4.1 Functional requirements

1. **Data Collection and Preprocessing**:
    - The system must collect and preprocess DMS past papers, lecture content, and syllabus from SLIIT, ensuring data is cleaned and structured for analysis by extracting text, removing noise, and organizing it into a usable format.
2. **Difficulty Prediction**:
    - The system shall predict essay question difficulty by analyzing lecture materials and past exam data, generating a multi-level difficulty index (e.g., Easy, Medium, Hard) based on content alignment to guide student preparation.
3. **Practice Question Access**:
    - The system must provide students with access to categorized past paper questions, allowing selection based on difficulty level to support targeted practice sessions.
4. **Essay Submission**:
    - The system shall allow students to submit essay responses for evaluation, supporting text input via a user-friendly interface with submission tracking and confirmation.
5. **Automated Evaluation**:
    - The system must evaluate submitted essays using NLP-based semantic similarity and Bloom's Taxonomy criteria, assessing quality, relevance, and completeness to provide a fair assessment.
6. **Feedback Generation**:
    - The system shall provide a personalized feedback, including suggested topics to revisit, to support student improvement and learning focus,optimized by reinforcement learning and AutoML.
7. **Adaptive Learning Loop**:
    - The system must adapt feedback and practice recommendations based on individual student performance trends over time to enhance personalized learning experiences.
8. **User Interface**:
    - The system must offer an intuitive interface with features like progress trackers, difficulty filters, and categorized question access for both students and educators to ensure ease of use.

3.4.2 Non-Functional Requirements

1. **Usability**:
    - The interface must be intuitive and accessible, achieving a high usability score in user testing with SLIIT students to ensure a positive user experience.
2. **Performance**:
    - The system must process and evaluate an essay submission within a reasonable timeframe under typical load conditions to maintain efficiency.
3. **Scalability**:
    - The system should handle an increase in users and data (e.g., additional modules or years) with minimal performance degradation to support future growth.
4. **Security**:
    - The system must ensure data privacy by encrypting student submissions and adhering to SLIIT data protection policies to safeguard sensitive information.
5. **Reliability**:
    - The system should maintain high uptime, with automated backups performed regularly to prevent data loss and ensure continuous availability.
6. **Compatibility**:
    - The system must be accessible on common web browsers and mobile devices without requiring additional plugins for broad accessibility.
7. **Maintainability**:
    - The codebase must be modular, with clear documentation, allowing updates or fixes within a reasonable timeframe by a single developer to facilitate ongoing development.
8. **Accuracy**:
    - The difficulty prediction and essay scoring models must achieve a strong accuracy level compared to human evaluations, validated through testing to ensure reliability.

3.5 Expected Test Cases

Test Case ID	Input	Action	Expected Output
TC-01	Upload past exam papers & lecture slides (PDF/DOCX/PPT	Preprocess (OCR, normalization, error correction)	Cleaned, structured dataset ready for mapping
TC-02	Input essay question	Run Difficulty Prediction Module (Transformer-based)	Multi-level difficulty index assigned to question
TC-03	Upload past paper question set	Map with lecture materials	Correct alignment between exam questions and lecture slides
TC-04	Student submits essay response	Automated Evaluation (NLP + Bloom's criteria)	completeness, and cognitive depth
TC-05	Student requests practice	System retrieves categorized past paper questions	Difficulty-tagged practice questions displayed
TC-06	Multiple students submit essays simultaneously	Validate modularity & concurrency	Each student receives independent evaluation and feedback
TC-07	Upload large dataset (10 years of past papers & lectures)	System scalability test	System processes without crash, maintaining performance
TC-08	User provides invalid input (e.g., blank essay or wrong file)	System validation	Error message displayed ("Invalid input, please resubmit")

3.6 Timeline

The proposed timeline for the project is as follows.

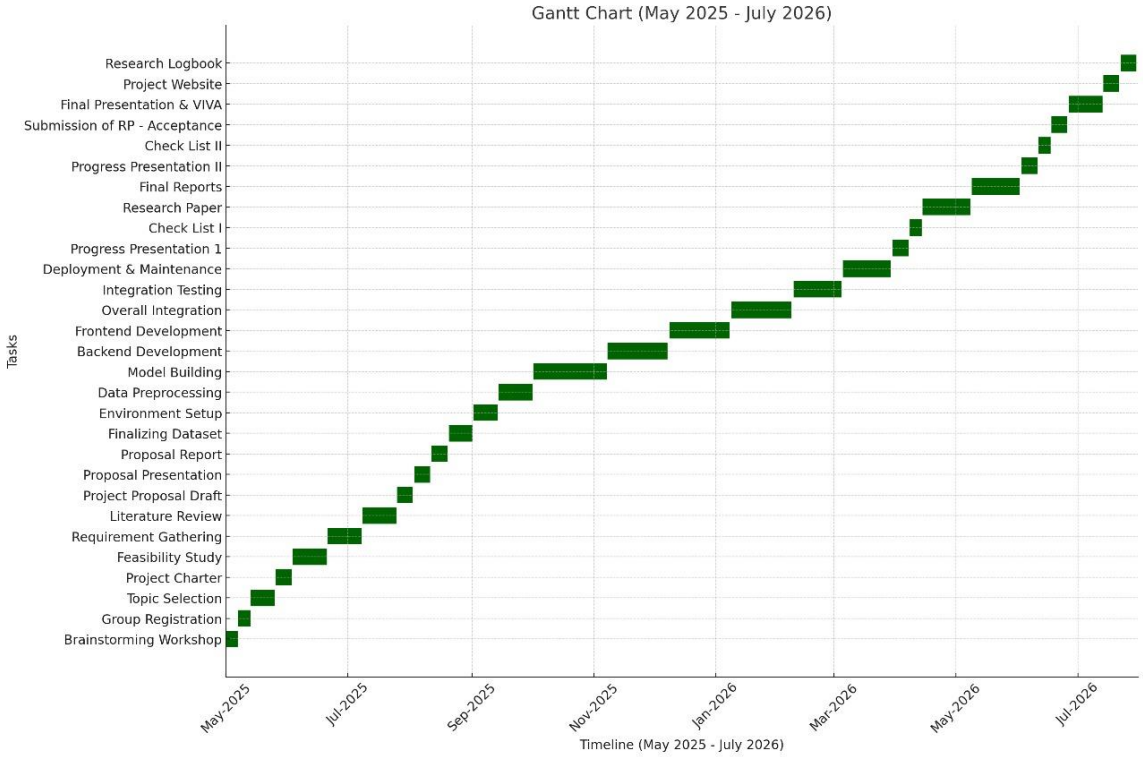

Figure 5: Gantt Chart for Project Schedule Management

3.7 Risk Management Plan

Risk	Description	Probability	Impact	Mitigation Strategy	Continency Plan
Technical	Model performance degradation due to limited training data.	Medium	Medium	Collect 8–10 years of historical DMS exam papers, lecture content, and CA questions; use data augmentation techniques.	Expand dataset with synthetic data; fine-tune transformer models
Operational	Inconsistent feedback generation from NLP models.	Medium	Medium	Implement rule-based feedback augmented with NLP insights; validate against expert evaluations.	Manually review and adjust feedback for critical cases.
Data-Related	Insufficient data quality affecting difficulty prediction accuracy.	Medium	High	Apply text extraction, normalization, tokenization, and semantic tagging; ensure data alignment with curriculum.	Conduct additional data cleaning and preprocessing cycles.
External	Syllabus changes affecting model relevance.	Low	Medium	Monitor updates via supervisor meetings; design modular system for retraining in Visual Studio Code.	Retrain models with updated syllabus data; adjust study plan algorithms.
External	Budget overruns for cloud resources (e.g., Colab/Kaggle).	Medium	Medium	Track costs monthly via Jira; prioritize free tiers of Colab/Kaggle and open-source tools.	Shift to local processing with Visual Studio Code; reduce model complexity.

4.0 COMMERCIALIZATION

The proposed "An Intelligent Essay-Based Learning Support System with Difficulty Prediction and Feedback-Driven Improvement" offers significant commercialization potential within the higher education and EdTech sectors, driven by the growing need for AI-enhanced, personalized learning tools that improve curriculum alignment and assessment preparation. Leveraging its strengths in advanced Natural Language Processing (NLP) techniques, reinforcement learning, AutoML transformer-based models like BERT, and a modular architecture, the system provides scalable, privacy-compliant solutions tailored to technical modules such as Database Management Systems (DMS) at institutions like the Sri Lanka Institute of Information Technology (SLIIT). Target markets include universities, colleges, online learning platforms, and tutoring services aiming to enhance student engagement and academic outcomes. The commercialization strategy could involve licensing the system to educational institutions or offering it as a subscription-based SaaS platform with features like automated essay scoring and educator insights. Pilot testing at SLIIT, as outlined in the methodology, will validate the system's efficacy, while partnerships with EdTech firms could facilitate broader adoption. Protecting intellectual property through patents and establishing robust technical support and training will be essential for market success, positioning the system to meet the evolving demands of educators and students effectively.

5.0 BUDGET & BUDGET JUSTIFICATION

Since the outcome of the proposed model is a software-based solution, there are no hardware components connected to the implementation. The primary source of the cost will be the subscription fees to the cloud provider for the computing power of the virtual machines. However, there will be some other costs expected to as be given in the table below.

Type	**Cost**
Internet use and web hosting	10 000 LKR
Training Cost	10 000 LKR
Publication Cost	50 000 LKR
Stationery	1 000 LKR
Additional	10 000 LKR
TOTAL	***81 000 LKR***

6.0 SUMMARY

In conclusion, the proposed "An Intelligent Essay-Based Learning Support System with Difficulty Prediction and Feedback-Driven Improvement" demonstrates its potential as a transformative academic tool for students in the DMS module at SLIIT by integrating advanced AI capabilities into a cohesive platform. Through the use of transformer-based models, the system effectively predicts essay difficulty and aligns it with lecture content, providing students with tailored practice opportunities. The inclusion of automated evaluation and personalized feedback enhances active learning, while the adaptive learning loop supports diverse student needs by adjusting recommendations based on performance trends. Furthermore, the curriculum alignment feature ensures that educators receive actionable insights to refine teaching strategies, improving overall curriculum coherence.

By delivering these features in a structured and user-friendly interface, the system not only boosts exam preparation efficiency but also fosters an inclusive learning experience that adapts to individual requirements. Its design balances technical innovation with educational rigor, ensuring reliability, relevance, and accessibility. Overall, this system lays a solid foundation for a more intelligent and personalized learning environment, significantly enhancing student outcomes and institutional effectiveness in the DMS module.

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